

Red Hat OpenShift Data Foundation 4.22

Deploying OpenShift Data Foundation on VMware vSphere

Instructions on deploying OpenShift Data Foundation using VMware vSphere infrastructure

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Abstract

Read this document for instructions about how to install Red Hat OpenShift Data Foundation using Red Hat OpenShift Container Platform on VMware vSphere clusters.

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PREFACE

This document describes how to deploy Red Hat OpenShift Data Foundation on VMware vSphere using dynamic or local storage devices.

PROVIDING FEEDBACK ON RED HAT DOCUMENTATION

We appreciate your input on our documentation. Do let us know how we can make it better.

To give feedback, create a Jira ticket:

Procedure

1. Log in to the Jira.
2. Click **Create** in the top navigation bar
3. Enter a descriptive title in the Summary field.
4. Enter your suggestion for improvement in the Description field. Include links to the relevant parts of the documentation.
5. Select **Documentation** in the Components field.
6. Click Create at the bottom of the dialogue.

PREFACE

Red Hat OpenShift Data Foundation supports deployment on existing Red Hat OpenShift Container Platform (RHOCP) vSphere clusters in connected or disconnected environments along with out-of-the-box support for proxy environments.

To deploy OpenShift Data Foundation, start with the requirements in the Preparing to deploy OpenShift Data Foundation chapter and then follow the appropriate deployment process:

- Internal mode — Deploy using dynamic storage devices, Deploy using local storage devices, or Deploy standalone Multicloud Object Gateway
- External mode — Deploying OpenShift Data Foundation in external mode

Additional resources

- Preparing to deploy OpenShift Data Foundation
- Deploying OpenShift Data Foundation in external mode

CHAPTER 1. PREPARING TO DEPLOY OPENSHIFT DATA FOUNDATION

Deploying OpenShift Data Foundation on OpenShift Container Platform using dynamic or local storage devices provides you with the option to create internal cluster resources. This will result in the internal provisioning of the base services, which helps to make additional storage classes available to applications.

Before you begin the deployment of Red Hat OpenShift Data Foundation using dynamic or local storage, ensure that your resource requirements are met. See the Resource requirements section in the Planning guide.

1. Verify the rotational flag on your VMDKs before deploying object storage devices (OSDs) on them. For more information, see the knowledgebase article Override device rotational flag in ODF environment.

2. Optional: If you want to enable cluster-wide encryption using the external Key Management System (KMS) HashiCorp Vault, follow these steps:

   - Ensure that you have a valid Red Hat OpenShift Data Foundation Advanced subscription. To know how subscriptions for OpenShift Data Foundation work, see knowledgebase article on OpenShift Data Foundation subscriptions.

   - When the Token authentication method is selected for encryption then refer to Enabling cluster-wide encryption with KMS using the Token authentication method.

   - When the Kubernetes authentication method is selected for encryption then refer to Enabling cluster-wide encryption with KMS using the Kubernetes authentication method .

   - Ensure that you are using signed certificates on your Vault servers.

3. Optional: If you want to enable cluster-wide encryption using the external Key Management System (KMS) Thales CipherTrust Manager, you must first enable the Key Management Interoperability Protocol (KMIP) and use signed certificates on your server. Follow these steps:

   a. Create a KMIP client if one does not exist. From the user interface, select **KMIP → Client Profile → Add Profile**.

      i. Add the **CipherTrust** username to the **Common Name** field during profile creation.

   b. Create a token by navigating to **KMIP → Registration Token → New Registration Token**. Copy the token for the next step.

   c. To register the client, navigate to **KMIP → Registered Clients → Add Client**. Specify the **Name**. Paste the **Registration Token** from the previous step, then click **Save**.

   d. Download the Private Key and Client Certificate by clicking **Save Private Key** and **Save Certificate** respectively.

   e. To create a new KMIP interface, navigate to **Admin Settings → Interfaces → Add Interface**.

      i. Select **KMIP Key Management Interoperability Protocol** and click **Next**.

      ii. Select a free **Port**.

      iii. Select **Network Interface** as **all**.

iv. Select **Interface Mode** as **TLS, verify client cert, user name taken from client cert, auth request is optional**.

v. (Optional) You can enable hard delete to delete both metadata and material when the key is deleted. It is disabled by default.

vi. Select the CA to be used, and click **Save**.

f. To get the server CA certificate, click on the Action menu **(⋮)** on the right of the newly created interface, and click **Download Certificate**.

g. Optional: If StorageClass encryption is to be enabled during deployment, create a key to act as the Key Encryption Key (KEK):

i. Navigate to **Keys → Add Key**.

ii. Enter **Key Name**.

iii. Set the **Algorithm** and **Size** to **AES** and **256** respectively.

iv. Enable **Create a key in Pre-Active state**and set the date and time for activation.

v. Ensure that **Encrypt** and **Decrypt** are enabled under **Key Usage**.

vi. Copy the ID of the newly created Key to be used as the Unique Identifier during deployment.

4. Minimum starting node requirements
An OpenShift Data Foundation cluster will be deployed with minimum configuration when the standard deployment resource requirement is not met. See Resource requirements section in Planning guide.

5. Disaster recovery requirements
Disaster Recovery features supported by Red Hat OpenShift Data Foundation require all of the following prerequisites to successfully implement a disaster recovery solution:

- A valid Red Hat OpenShift Data Foundation Advanced subscription
- A valid Red Hat Advanced Cluster Management for Kubernetes subscription
To know how subscriptions for OpenShift Data Foundation work, see knowledgebase article on OpenShift Data Foundation subscriptions.

For detailed requirements, see Configuring OpenShift Data Foundation Disaster Recovery for OpenShift Workloads guide, and *Requirements and recommendations* section of the Install guide in Red Hat Advanced Cluster Management for Kubernetes documentation.

6. For deploying using local storage devices, see requirements for installing OpenShift Data Foundation using local storage devices. These are not applicable for deployment using dynamic storage devices.

7. For VMs on VMware, ensure the **disk.EnableUUID** option is set to **TRUE**. You need to have vCenter account privileges to configure the VMs. For more information, see Required vCenter account privileges. To set the **disk.EnableUUID** option, use the **Advanced** option of the **VM Options** in the **Customize hardware tab**. For more information, see Installing on vSphere.
Optional: If you want to use thick-provisioned storage for flexibility, you must create a storage class with **zeroedthick** or **eagerzeroedthick** disk format. For information, see VMware vSphere object definition.

1.1. RECOMMENDED CONFIGURATION FOR OPTIMIZED PERFORMANCE

Configure VMware vSphere storage settings to optimize OpenShift Data Foundation performance for small block I/O workloads such as databases that require low-latency disk flushes.

When deploying OpenShift Data Foundation to provide storage for workloads, it is important to consider applications that write small block size I/Os (4K-16K) and require the data to be flushed to disk before acknowledgement back to the client.

A common example of such a workload is a database workload.

In these scenarios, it is critical to ensure the storage presented to VMware limits the amount of layers between OpenShift Data Foundation and the actual backing physical device providing the persistent volume (PV) storage for the OSD.

Due to the way OpenShift Data Foundation handles write operations, an operation that requires the output to be flushed to disk introduces additional latency. When a write is received by the primary OSD, it must first be committed to disk. Only then can OpenShift Data Foundation synchronously send two additional writes to the peer OSDs, which must also commit to disk before the write is acknowledged back to the client.

Write operations can be further delayed depending on the customer's infrastructure and platform. For instance, if the storage system includes both a virtualization layer and a RAID layer, each further delays the completion of the operation. These stacked layers significantly impact performance, especially for small block (4K-16K) I/O that require the write to be flushed to disk before being acknowledged.

Based on the I/O flow just described, deploying OpenShift Data Foundation on VMware vSAN is **not** recommended when the backing storage device being presented into the datastore for vSAN uses RAID.

1.1.1. Recommended configuration guidelines

- **Use Passthrough NVMe disks**: The optimal setup for OpenShift Data Foundation on VMware is to configure it with passthrough NVMe disks. This bypasses the vSAN layer, thereby avoiding the additional latency it introduces.
- **Leverage LSO for device consumption**: Passthrough devices should be consumed using Local Storage Operator (LSO).
- **Avoid Default vSAN StorageClass for OSDs** In IPI deployments of OpenShift on vSAN, a default StorageClass **thin-csi** is automatically created. Do not use this StorageClass for OSDs, as it relies on RAID-1, which adds unnecessary overhead. Instead, use a StorageClass configured in thick lazy zeroed format.
- **Do not share storage with other workloads** Sharing storage between OpenShift Data Foundation and other workloads can significantly degrade performance. A dedicated StorageClass should be used exclusively for OpenShift Data Foundation.
- **Isolate ODF Storage Network Traffic** To prevent network congestion, it is recommended to use an isolated VM network for OpenShift Data Foundation storage traffic.

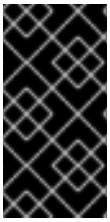
Sample impacted workloads:

- MongoDB

- SQLDB
- Cloud Pak for Data
- Cloud Pak for AI

Remediation steps

If your deployment currently uses a configuration similar to the one described above, we recommend reviewing the Knowledge Base article for detailed guidance on migrating from a vSAN-based setup to a passthrough device configuration.



IMPORTANT

If your VMware deployment matches the configuration described in the introduction of this section and you choose not to migrate to a setup that minimizes the layers between the Software-Defined Storage (SDS) and the physical storage, your environment may not meet the I/O performance requirements of certain applications.

Additional resources

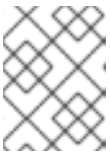
- Knowledge Base article: Migrating from vSAN-based setup to passthrough device configuration

1.2. REQUIREMENTS FOR INSTALLING OPENSHIFT DATA FOUNDATION USING LOCAL STORAGE DEVICES

Before installing OpenShift Data Foundation using local storage devices, your cluster must meet certain requirements.

Node requirements The cluster must consist of at least three OpenShift Container Platform worker or infrastructure nodes with locally attached-storage devices on each of them.

- Each of the three selected nodes must have at least one raw block device available. OpenShift Data Foundation uses the one or more available raw block devices.



NOTE

Make sure that the devices have a unique **by-id** device name for each available raw block device.

- The devices you use must be empty, the disks must not include Physical Volumes (PVs), Volume Groups (VGs), or Logical Volumes (LVs) remaining on the disk.

For more information, see the *Resource requirements* section in the Planning guide.

Disaster recovery requirements

Disaster Recovery features supported by Red Hat OpenShift Data Foundation require all of the following prerequisites to successfully implement a disaster recovery solution:

- A valid Red Hat OpenShift Data Foundation Advanced subscription.
- A valid Red Hat Advanced Cluster Management (RHACM) for Kubernetes subscription.

To know in detail how subscriptions for OpenShift Data Foundation work, see the knowledgebase article on OpenShift Data Foundation subscriptions.

For detailed disaster recovery solution requirements, see the *Configuring OpenShift Data Foundation Disaster Recovery for OpenShift Workloads* guide, and *Requirements and recommendations* section of the Install guide in Red Hat Advanced Cluster Management for Kubernetes documentation.

Arbiter stretch cluster requirements

In this case, a single cluster is stretched across two zones with a third zone as the location for the arbiter.

To know in detail how subscriptions for OpenShift Data Foundation work, see the knowledgebase article on OpenShift Data Foundation subscriptions.



NOTE

You cannot enable Flexible scaling and Arbiter both at the same time as they have conflicting scaling logic. With Flexible scaling, you can add one node at a time to your OpenShift Data Foundation cluster. Whereas, in an Arbiter cluster, you need to add at least one node in each of the two data zones.

Minimum starting node requirements

An OpenShift Data Foundation cluster is deployed with a minimum configuration when the resource requirement for a standard deployment is not met.

For more information, see the *Resource requirements* section in the Planning guide.

Disconnected environment requirements

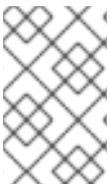
For the list of packages to be mirrored for installing OpenShift Data Foundation in disconnected environments, see the Disconnected environment section in the *Planning your deployment* guide.

Additional resources

- Planning guide - Disconnected environment
- Planning guide - Resource requirements
- Knowledgebase article on OpenShift Data Foundation subscriptions
- Configuring OpenShift Data Foundation Disaster Recovery for OpenShift Workloads
- Red Hat Advanced Cluster Management for Kubernetes - Requirements and recommendations
- Knowledgebase article on OpenShift Data Foundation subscriptions

CHAPTER 2. DEPLOY USING DYNAMIC STORAGE DEVICES

Deploying OpenShift Data Foundation on OpenShift Container Platform using dynamic storage devices provided by VMware vSphere (disk format: thin) provides you with the option to create internal cluster resources. This will result in the internal provisioning of the base services, which helps to make additional storage classes available to applications.



NOTE

Both internal and external OpenShift Data Foundation clusters are supported on VMware vSphere. See Planning your deployment for more information about deployment requirements.

Also, ensure that you have addressed the requirements in Preparing to deploy OpenShift Data Foundation chapter before proceeding with the below steps for deploying using dynamic storage devices:

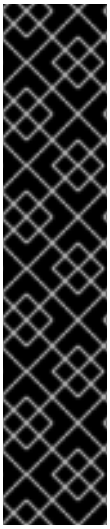
1. Install the Red Hat OpenShift Data Foundation Operator .
2. Create an OpenShift Data Foundation Cluster .

2.1. INSTALLING RED HAT OPENSHIFT DATA FOUNDATION OPERATOR

You can install Red Hat OpenShift Data Foundation Operator using the Red Hat OpenShift Container Platform Software Catalog.

Prerequisites

- Access to an OpenShift Container Platform cluster using an account with **cluster-admin** and operator installation permissions.
- You must have at least three worker or infrastructure nodes in the Red Hat OpenShift Container Platform cluster.
- For additional resource requirements, see the Planning your deployment guide.



IMPORTANT

- When you need to override the cluster-wide default node selector for OpenShift Data Foundation, you can use the following command to specify a blank node selector for the **openshift-storage** namespace (create **openshift-storage** namespace in this case):

```
$ oc annotate namespace openshift-storage openshift.io/node-selector=
```

- Taint a node as **infra** to ensure only Red Hat OpenShift Data Foundation resources are scheduled on that node. This helps you save on subscription costs. For more information, see the *How to use dedicated worker nodes for Red Hat OpenShift Data Foundation* section in the Managing and Allocating Storage Resources guide.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Ecosystem → Software Catalog**
3. Scroll or type **OpenShift Data Foundation** into the **Filter by keyword** box to find the **OpenShift Data Foundation** Operator.
4. Click **Install**.
5. Set the following options on the **Install Operator** page:
    a. Update Channel as **stable-4.22**.
    b. Installation Mode as **A specific namespace on the cluster**
    c. Installed Namespace as **Operator recommended namespace openshift-storage**. If Namespace **openshift-storage** does not exist, it is created during the operator installation.
    d. Select Approval Strategy as **Automatic** or **Manual**.
    If you select **Automatic** updates, then the Operator Lifecycle Manager (OLM) automatically upgrades the running instance of your Operator without any intervention.

    If you select **Manual** updates, then the OLM creates an update request. As a cluster administrator, you must then manually approve that update request to update the Operator to a newer version.
    e. Ensure that the **Enable** option is selected for the **Console plugin**.
    f. Click **Install**.

Verification

- After the operator is successfully installed, the web console automatically reloads to apply the changes. During this process, a temporary error message might appear on the page and this is expected and disappears after the refresh completes.
- In the Web Console:
    - Navigate to Installed Operators and verify that the **OpenShift Data Foundation** Operator shows a green tick indicating successful installation.
    - Navigate to **Storage** and verify if the **Data Foundation** dashboard is available.

2.2. ENABLING CLUSTER-WIDE ENCRYPTION WITH KMS USING THE TOKEN AUTHENTICATION METHOD

You can enable the key value backend path and policy in the vault for token authentication.

Prerequisites

- Administrator access to the vault.
- A valid Red Hat OpenShift Data Foundation Advanced subscription. For more information, see the knowledgebase article on OpenShift Data Foundation subscriptions.

- Carefully, select a unique path name as the backend **path** that follows the naming convention since you cannot change it later.

Procedure

1. Enable the Key/Value (KV) backend path in the vault.
   For vault KV secret engine API, version 1:

   ```
   $ vault secrets enable -path=odf kv
   ```

   For vault KV secret engine API, version 2:

   ```
   $ vault secrets enable -path=odf kv-v2
   ```

2. Create a policy to restrict the users to perform a write or delete operation on the secret:

   ```
   echo '
   path "odf/*" {
     capabilities = ["create", "read", "update", "delete", "list"]
   }
   path "sys/mounts" {
   capabilities = ["read"]
   }'| vault policy write odf -
   ```

3. Create a token that matches the above policy:

   ```
   $ vault token create -policy=odf -format json
   ```

2.3. ENABLING CLUSTER-WIDE ENCRYPTION WITH KMS USING THE KUBERNETES AUTHENTICATION METHOD

You can enable the Kubernetes authentication method for cluster-wide encryption using the Key Management System (KMS).

Prerequisites

- Administrator access to Vault.
- A valid Red Hat OpenShift Data Foundation Advanced subscription. For more information, see the knowledgebase article on OpenShift Data Foundation subscriptions.
- The OpenShift Data Foundation operator must be installed from the Software Catalog.
- Select a unique path name as the backend **path** that follows the naming convention carefully. You cannot change this path name later.

Procedure

1. Create a service account:

   ```
   $ oc -n openshift-storage create serviceaccount <serviceaccount_name>
   ```

   where, ***<serviceaccount_name>*** specifies the name of the service account.

For example:

```
$ oc -n openshift-storage create serviceaccount odf-vault-auth
```

2. Create **clusterrolebindings** and **clusterroles**:

```
$ oc -n openshift-storage create clusterrolebinding vault-tokenreview-binding --
clusterrole=system:auth-delegator --serviceaccount=openshift-
storage:_<serviceaccount_name>_
```

For example:

```
$ oc -n openshift-storage create clusterrolebinding vault-tokenreview-binding --
clusterrole=system:auth-delegator --serviceaccount=openshift-storage:odf-vault-auth
```

3. Create a secret for the **serviceaccount** token and CA certificate.

```
$ cat <<EOF | oc create -f -
apiVersion: v1
kind: Secret
metadata:
  name: odf-vault-auth-token
  namespace: openshift-storage
  annotations:
    kubernetes.io/service-account.name: <serviceaccount_name>
type: kubernetes.io/service-account-token
data: {}
EOF
```

where, ***<serviceaccount_name>*** is the service account created in the earlier step.

4. Get the token and the CA certificate from the secret.

```
$ SA_JWT_TOKEN=$(oc -n openshift-storage get secret odf-vault-auth-token -o jsonpath="
{.data['token']}" | base64 --decode; echo)
$ SA_CA_CRT=$(oc -n openshift-storage get secret odf-vault-auth-token -o jsonpath="
{.data['ca\.crt']}" | base64 --decode; echo)
```

5. Retrieve the OCP cluster endpoint.

```
$ OCP_HOST=$(oc config view --minify --flatten -o jsonpath="{.clusters[0].cluster.server}")
```

6. Fetch the service account issuer:

```
$ oc proxy &
$ proxy_pid=$!
$ issuer="$( curl --silent http://127.0.0.1:8001/.well-known/openid-configuration | jq -r
.issuer)"
$ AUDIENCE=$(echo '{"apiVersion": "authentication.k8s.io/v1", "kind": "TokenRequest"}' |
kubectl create f --raw /api/v1/namespaces/default/serviceaccounts/default/token | jq -r
'.spec.audiences' | jq -r .[0])
$ kill $proxy_pid
```

Or, for Vault version 1.21.0 and higher:

```
$ oc proxy &
$ proxy_pid=$!
$ ISSUER=$(echo '{"apiVersion": "authentication.k8s.io/v1", "kind": "TokenRequest"}' \   |
kubectl create f --raw /api/v1/namespaces/default/serviceaccounts/default/token \   | jq -r
'.status.token' \   | cut -d . -f2 \   | base64 -d \   | jq '.iss')
$ AUDIENCE=$(echo '{"apiVersion": "authentication.k8s.io/v1", "kind": "TokenRequest"}' |
kubectl create f --raw /api/v1/namespaces/default/serviceaccounts/default/token | jq -r
'.spec.audiences' | jq -r .[0])
$ kill $proxy_pid
```

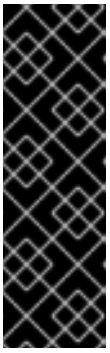
7. Use the information collected in the previous step to setup the Kubernetes authentication method in Vault:

```
$ vault auth enable kubernetes
```

```
$ vault write auth/kubernetes/config \
        token_reviewer_jwt="$SA_JWT_TOKEN" \
        kubernetes_host="$OCP_HOST" \
        kubernetes_ca_cert="$SA_CA_CRT" \
        issuer="$issuer"
```

Or, for Vault version 1.21.0 and higher:

```
$ oc exec -n vault -ti vault-0 – \
   vault write auth/kubernetes/config \
   token_reviewer_jwt="$SA_JWT_TOKEN" \
   kubernetes_host="$OCP_HOST" \
   kubernetes_ca_cert="$SA_CA_CRT" \
   issuer="$ISSUER"\
   token_audience="$AUDIENCE"
```



IMPORTANT

To configure the Kubernetes authentication method in Vault when the issuer is empty:

```
$ vault write auth/kubernetes/config \
        token_reviewer_jwt="$SA_JWT_TOKEN" \
        kubernetes_host="$OCP_HOST" \
        kubernetes_ca_cert="$SA_CA_CRT"
```

8. Enable the Key/Value (KV) backend path in Vault.
For Vault KV secret engine API, version 1:

```
$ vault secrets enable -path=odf kv
```

For Vault KV secret engine API, version 2:

```
$ vault secrets enable -path=odf kv-v2
```

9. Create a policy to restrict the users to perform a **write** or **delete** operation on the secret:

```
echo '
path "odf/*" {
  capabilities = ["create", "read", "update", "delete", "list"]
}
path "sys/mounts" {
capabilities = ["read"]
}'| vault policy write odf -
```

10. Generate the roles:

```
$ vault write auth/kubernetes/role/odf-rook-ceph-op \
        bound_service_account_names=rook-ceph-system,rook-ceph-osd,noobaa \
        bound_service_account_namespaces=openshift-storage \
        policies=odf \
        ttl=1440h
```

The role **odf-rook-ceph-op** is later used while you configure the KMS connection details during the creation of the storage system.

```
$ vault write auth/kubernetes/role/odf-rook-ceph-osd \
        bound_service_account_names=rook-ceph-osd \
        bound_service_account_namespaces=openshift-storage \
        policies=odf \
        ttl=1440h
```

2.3.1. Enabling and disabling key rotation when using KMS

Security common practices require periodic encryption of key rotation. You can enable or disable key rotation when using KMS.

2.3.1.1. Enforcing precedence for key rotation

Configure key rotation precedence to control whether the rotation schedule is read from the PVC annotation or the storage class in OpenShift Data Foundation.

In key rotation, precedence refers to the order in which the system checks for scheduled annotations. In OpenShift Data Foundation, the default precedence is set to storage class (recommended). This means the system reads annotations only from the storage class.

However, if you want the system to check the persistent volume claim (PVC) first and then fall back to the storage class, you can configure this behavior by setting the **schedule-precedence** to PVC in the CSI-addons ConfigMap.

Procedure

1. Define the **ConfigMap** for csi-addons to set PVC precedence:

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: csi-addons-config
```

```
  namespace: openshift-storage
data:
  "schedule-precedence": "pvc"
```

2. Restart the **csi-addons** operator pod for the changes to take effect:

```
oc delete po -n openshift-storage -l "app.kubernetes.io/name=csi-addons"
```

2.3.1.2. Enabling key rotation

Enable key rotation to automatically re-encrypt persistent volume data at a scheduled interval using the CSI Addons framework in OpenShift Data Foundation.

To enable key rotation, add the annotation **keyrotation.csiaddons.openshift.io/schedule: <value>** to a **PersistentVolumeClaim**, **Namespace**, or **StorageClass** (in decreasing order of precedence).



IMPORTANT

Key rotation is only supported for RBD backed volumes.

2.3.1.2.1. Annotating a namespace for key rotation

Add the key rotation annotation to a namespace so that all RBD-backed persistent volumes in that namespace are re-encrypted on the configured schedule.

Prerequisites

- Key rotation is only supported for RBD backed volumes.

Procedure

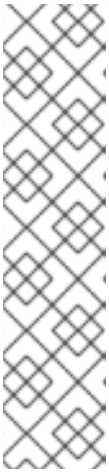
1. Verify the namespace exists:

```
$ oc get namespace default
NAME      STATUS   AGE
default   Active   5d2h
```

2. Annotate the namespace with the desired rotation schedule:

```
$ oc annotate namespace default "keyrotation.csiaddons.openshift.io/schedule=@weekly"
namespace/default annotated
```

<value> can be **@hourly**, **@daily**, **@weekly**, **@monthly**, or **@yearly**. If **<value>** is empty, the default is **@weekly**.



NOTE

- When any schedule for key rotation is used, including **@daily**, **@weekly**, **@monthly**, and cron expressions such as **0 * * * ***, multiple CronJobs are created. To prevent all Jobs from running simultaneously and overloading high-scale systems, executions are staggered within a 2-hour window from the scheduled start time (e.g., **@weekly** would be 00:00, 00:05, 00:10, and so on).
    - If the configured schedule interval is less than 2 hours, the stagger window is reduced to match the interval. For example, a job running every 30 minutes will be staggered within a 00:00–00:30 window rather than 00:00–02:00.

2.3.1.2.2. Annotating a storage class for key rotation

Add the key rotation annotation to a storage class so that all RBD-backed persistent volumes provisioned from that storage class are re-encrypted on the configured schedule.

Prerequisites

- Key rotation is only supported for RBD backed volumes.

Procedure

1. Verify the storage class exists:

```
$ oc get storageclass rbd-sc
NAME      PROVISIONER        RECLAIMPOLICY   VOLUMEBINDINGMODE
ALLOWVOLUMEEXPANSION   AGE
rbd-sc    rbd.csi.ceph.com   Delete          Immediate           true                   5d2h
```

2. Annotate the storage class with the desired rotation schedule:

```
$ oc annotate storageclass rbd-sc "keyrotation.csiaddons.openshift.io/schedule=@weekly"
storageclass.storage.k8s.io/rbd-sc annotated
```

<value> can be **@hourly**, **@daily**, **@weekly**, **@monthly**, or **@yearly**. If **<value>** is empty, the default is **@weekly**.

2.3.1.2.3. Annotating a PersistentVolumeClaim for key rotation

Add the key rotation annotation directly to a PersistentVolumeClaim to re-encrypt that volume on the configured schedule, overriding any namespace or storage class annotation.

Prerequisites

- Key rotation is only supported for RBD backed volumes.

Procedure

1. Verify the PersistentVolumeClaim exists:

```
$ oc get pvc data-pvc
```

```
NAME      STATUS   VOLUME                                     CAPACITY   ACCESS MODES
STORAGECLASS      AGE
data-pvc  Bound    pvc-f37b8582-4b04-4676-88dd-e1b95c6abf74   1Gi        RWO
default           20h
```

2. Annotate the PVC with the desired rotation schedule:

```
$ oc annotate pvc data-pvc "keyrotation.csiaddons.openshift.io/schedule=@weekly"
persistentvolumeclaim/data-pvc annotated
```

<value> can be **@hourly**, **@daily**, **@weekly**, **@monthly**, or **@yearly**. If **<value>** is empty, the default is **@weekly**.

3. Verify the encryption key rotation cron job was created:

```
$ oc get encryptionkeyrotationcronjobs.csiaddons.openshift.io
NAME                  SCHEDULE    SUSPEND   ACTIVE   LASTSCHEDULE   AGE
data-pvc-1642663516   @weekly                                       3s
```

4. Optional: Update the schedule by overwriting the annotation:

```
$ oc annotate pvc data-pvc "keyrotation.csiaddons.openshift.io/schedule=*/1 * * * *" --
overwrite=true
persistentvolumeclaim/data-pvc annotated
```

5. Verify the updated cron job schedule:

```
$ oc get encryptionkeyrotationcronjobs.csiaddons.openshift.io
NAME                  SCHEDULE    SUSPEND   ACTIVE   LASTSCHEDULE   AGE
data-pvc-1642664617   */1 * * * *                                   3s
```

2.3.1.3. Disabling key rotation

Disable automatic key rotation for all PVCs in a storage class or for a specific persistent volume claim in OpenShift Data Foundation.

You can disable key rotation for the following:

- All the persistent volume claims (PVCs) of storage class
- A specific PVC

Disabling key rotation for all PVCs of a storage class

To disable key rotation for all PVCs, update the annotation of the storage class:

```
$ oc get storageclass rbd-sc
NAME      PROVISIONER        RECLAIMPOLICY   VOLUMEBINDINGMODE
ALLOWVOLUMEEXPANSION   AGE
rbd-sc    rbd.csi.ceph.com   Delete          Immediate           true                   5d2h
```

```
$ oc annotate storageclass rbd-sc "keyrotation.csiaddons.openshift.io/enable: false"
storageclass.storage.k8s.io/rbd-sc annotated
```

Disabling key rotation for a specific persistent volume claim

Procedure

1. Identify the **EncryptionKeyRotationCronJob** CR for the PVC you want to disable key rotation on:

```
$ oc get encryptionkeyrotationcronjob -o jsonpath='{range .items[?
(@.spec.jobTemplate.spec.target.persistentVolumeClaim=="<PVC_NAME>")]}
{.metadata.name}{"\n"}{end}'
```

   Where **<PVC_NAME>** is the name of the PVC that you want to disable.

2. Apply the following to the **EncryptionKeyRotationCronJob** CR from the previous step to disable the key rotation:

   a. Update the **csiaddons.openshift.io/state** annotation from **managed** to **unmanaged**:

```
$ oc annotate encryptionkeyrotationcronjob <encryptionkeyrotationcronjob_name>
"csiaddons.openshift.io/state=unmanaged" --overwrite=true
```

   Where **<encryptionkeyrotationcronjob_name>** is the name of the **EncryptionKeyRotationCronJob** CR.

   b. Add **suspend: true** under the **spec** field:

```
$ oc patch encryptionkeyrotationcronjob <encryptionkeyrotationcronjob_name> -p
'{"spec": {"suspend": true}}' --type=merge.
```

   The key rotation will be disabled for the PVC.

2.4. CREATING OPENSHIFT DATA FOUNDATION CLUSTER

Create an OpenShift Data Foundation cluster after you install the OpenShift Data Foundation operator.

Prerequisites

- The OpenShift Data Foundation operator must be installed from the Software Catalog. For more information, see Installing OpenShift Data Foundation Operator .
- For VMs on VMware, if you have not already enabled UUIDs, ensure the **disk.EnableUUID** option is set to **TRUE**. You need to have vCenter account privileges to configure the VMs. For more information, see Required vCenter account privileges . To set the **disk.EnableUUID** option, use the **Advanced** option of the **VM Options** in the **Customize hardware tab**. For more information, see Installing on vSphere .
- Optional: If you want to use thick-provisioned storage for flexibility, you must create a storage class with **zeroedthick** or **eagerzeroedthick** disk format. For information, see VMware vSphere object definition.

Procedure

1. In the OpenShift Web Console, click **Storage → Data Foundation → Storage Systems → Create StorageSystem**.

2. In the **Backing storage** page, select the following:

   a. Select the **Use an existing StorageClass** option.

   b. Select the **Storage Class**.
   By default, it is set to **thin-csi**. If you have created a storage class with **zeroedthick** or **eagerzeroedthick** disk format for thick-provisioned storage, then that storage class is listed in addition to the default, **thin-csi** storage class.

   c. Click **Next**.

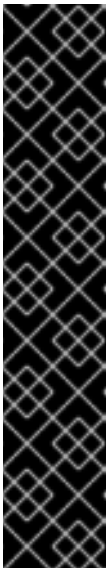
3. In the **Advanced Settings** page, you can configure the following optional items:

   - **Enable Network Files System**
     Automatically enable NFS for the cluster

   - **Use Ceph RBD as the default StorageClass**
     Sets Ceph RBD as the default StorageClass, eliminating the need to manually annotate a StorageClass.

   - **Set default StorageClass for virtualization**
     Marks the RBD virtualization storage class as the default for KubeVirt VM disks (PVs) after installation.

   - **Use external PostgreSQL** [Technology preview]
     Enables the use of an external PostgreSQL instance. This provides a high-availability solution for the Multicloud Object Gateway, where the PostgreSQL pod is a single point of failure.



IMPORTANT

OpenShift Data Foundation ships PostgreSQL images maintained by Red Hat, which are used to store metadata for the Multicloud Object Gateway. This PostgreSQL usage is at the application level.

As a result, OpenShift Data Foundation does not perform database-level optimizations or in-depth insights.

If you have well-maintained and optimized PostgreSQL instance, then it is recommended to use it. OpenShift Data Foundation supports external PostgreSQL instances.

Any PostgreSQL-related issues requiring code changes or deep technical analysis may need to be addressed upstream. This could result in longer resolution times.

   a. Provide the following connection details:

      - **Username**
      - **Password**
      - **Server name** and **Port**
      - **Database name**

b. Select **Enable TLS/SSL** checkbox to enable encryption for the Postgres server.

- **Enable automatic backup**
  Allows automatic backup for the Multicloud Object Gateway metadata database.

  a. Select the backup frequency: Daily, Weekly, or Monthly.

  b. Specify the number of backups to retain.



NOTE

This option is disabled if the **Use external PostgreSQL** option is selected.

a. Click **Next**.

1. In the **Capacity and nodes** page, provide the necessary information:

b. Select a value for **Requested Capacity** from the dropdown list. It is set to **2 TiB** by default.



NOTE

Once you select the initial storage capacity, cluster expansion is performed only using the selected usable capacity (three times of raw storage).

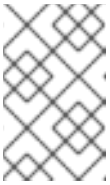
c. In the **Select Nodes** section, select at least three available nodes.

d. In the **Configure performance** section, select one of the following performance profiles:

- Lean
  Use this in a resource constrained environment with minimum resources that are lower than the recommended. This profile minimizes resource consumption by allocating fewer CPUs and less memory.

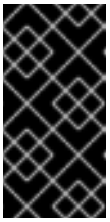
- Balanced (default)
  Use this when recommended resources are available. This profile provides a balance between resource consumption and performance for diverse workloads.

- Performance
  Use this in an environment with sufficient resources to get the best performance. This profile is tailored for high performance by allocating ample memory and CPUs to ensure optimal execution of demanding workloads.



NOTE

You have the option to configure the performance profile even after the deployment using the **Configure performance** option from the options menu of the **StorageSystems** tab.



IMPORTANT

Before selecting a resource profile, make sure to check the current availability of resources within the cluster. Opting for a higher resource profile in a cluster with insufficient resources might lead to installation failures.

For more information about resource requirements, see Resource requirement for performance profiles.

e. Optional: Select the **Taint nodes** checkbox to dedicate the selected nodes for OpenShift Data Foundation.
Spread the worker nodes across three different physical nodes, racks, or failure domains for high availability.

Use vCenter anti-affinity to align OpenShift Data Foundation rack labels with physical nodes and racks in the data center to avoid scheduling two worker nodes on the same physical chassis.

If the nodes selected do not match the OpenShift Data Foundation cluster requirement of the aggregated 30 CPUs and 72 GiB of RAM, a minimal cluster is deployed. For minimum starting node requirements, see the Resource requirements section in the Planning guide.

Select the **Taint nodes** checkbox to make selected nodes dedicated for OpenShift Data Foundation.

f. Optional: Select the **Enable automatic capacity scaling for your cluster**checkbox.
When automatic capacity scaling is enabled, additional raw capacity equivalent to the configured deployment size is automatically added to the cluster when used capacity reaches 70%. This ensures your deployment scales seamlessly to meet demand.

This option is disabled in lean profile mode, LSO deployment, and external mode deployment.



IMPORTANT

This may incur additional costs for the underlying storage.

i. Set the cluster expansion limit from the dropdown. This is the maximum the cluster can expand in the cloud. Automatic scaling is suspended if this limit is exceeded.

g. Click **Next**.

1. Optional: In the **Security and network** page, configure the following based on your requirements:

h. To enable encryption, select **Enable data encryption for block and file storage**

i. Select either one or both the encryption levels:

- **Cluster-wide encryption**
Encrypts the entire cluster (block and file).

- **StorageClass encryption**
Creates encrypted persistent volume (block only) using encryption enabled storage class.

ii. Optional: Select the **Connect to an external key management service** checkbox. This is optional for cluster-wide encryption.

A. From the **Key Management Service Provider** drop-down list, select one of the following providers and provide the necessary details:

- **Vault**

I. Select an **Authentication Method**.

- **Using Token authentication method**
  - Enter a unique **Connection Name**, host **Address** of the Vault server ('https://<hostname or ip>'), **Port** number and **Token**.
  - Expand **Advanced Settings** to enter additional settings and certificate details based on your **Vault** configuration:
    - Enter the Key Value secret path in **Backend Path** that is dedicated and unique to OpenShift Data Foundation.
  - Optional: Enter **TLS Server Name**, **Authentication Path**, and **Vault Enterprise Namespace**.
  - Upload the respective PEM encoded certificate file to provide the **CA Certificate**, **Client Certificate** and **Client Private Key**.
  - Click **Save**.
- **Using Kubernetes authentication method**
  - Enter a unique Vault **Connection Name**, host **Address** of the Vault server ('https://<hostname or ip>'), **Port** number and **Role** name.
  - Expand **Advanced Settings** to enter additional settings and certificate details based on your **Vault** configuration:
    - Enter the Key Value secret path in **Backend Path** that is dedicated and unique to OpenShift Data Foundation.
    - Optional: Enter **TLS Server Name**, **Authentication Path**, and **Vault Enterprise Namespace** if applicable.
    - Upload the respective PEM encoded certificate file to provide the **CA Certificate**, **Client Certificate** and **Client Private Key** .
  - Click **Save**.



NOTE

In case you need to enable key rotation for Vault KMS, run the following command in the OpenShift web console after the storage cluster is created:

```
$ oc patch storagecluster ocs-storagecluster -n openshift-storage --type=json -p '[{"op": "add", "path":"/spec/encryption/keyRotation/enable", "value": true}]'
```

- **Thales CipherTrust Manager (using KMIP)**

  I. Enter a unique **Connection Name** for the Key Management service within the project.

  II. In the **Address** and **Port** sections, enter the IP of Thales CipherTrust Manager and the port where the KMIP interface is enabled. For example:

  - **Address**: 123.34.3.2
  - **Port**: 5696

  III. Upload the **Client Certificate**, **CA certificate**, and **Client Private Key**.

  IV. If StorageClass encryption is enabled, enter the Unique Identifier to be used for encryption and decryption generated above.

  V. The **TLS Server** field is optional and used when there is no DNS entry for the KMIP endpoint. For example, **kmip_all_<port>.ciphertrustmanager.local**.

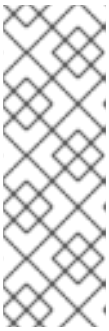
i. To enable in-transit encryption, select **In-transit encryption**.

i. Select a **Network**.

ii. Click **Next**.

1. In the **Review and create** page, review the configuration details.
   To modify any configuration settings, click **Back**.

2. Click **Create StorageSystem**.



NOTE

When your deployment has five or more nodes, racks, or rooms, and when there are five or more number of failure domains present in the deployment, you can configure Ceph monitor counts based on the number of racks or zones. An alert is displayed in the notification panel or Alert Center of the OpenShift Web Console to indicate the option to increase the number of Ceph monitor counts. You can use the **Configure** option in the alert to configure the Ceph monitor counts. For more information, see Resolving low Ceph monitor count alert .

Verification

- To verify the final Status of the installed storage cluster:

  a. In the OpenShift Web Console, navigate to **Storage → Data Foundation → Storage System → ocs-storagecluster**.

  b. Verify that **Status** of **StorageCluster** is **Ready** and has a green tick mark next to it.

  1. To verify that all components for OpenShift Data Foundation are successfully installed, see Verifying your OpenShift Data Foundation deployment .

Additional resources

- Alerts

CHAPTER 3. DEPLOY USING LOCAL STORAGE DEVICES

Deploying OpenShift Data Foundation on OpenShift Container Platform using local storage devices provides you with the option to create internal cluster resources. This will result in the internal provisioning of the base services, which helps to make additional storage classes available to applications.

Use this section to deploy OpenShift Data Foundation on VMware where OpenShift Container Platform is already installed.

Also, ensure that you have addressed the requirements in Preparing to deploy OpenShift Data Foundation chapter before proceeding with the next steps.

1. Installing Local Storage Operator
2. Install the Red Hat OpenShift Data Foundation Operator .
3. Create the OpenShift Data Foundation Cluster .

3.1. INSTALLING LOCAL STORAGE OPERATOR

Install the Local Storage Operator from the Software Catalog before creating Red Hat OpenShift Data Foundation clusters on local storage devices.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Ecosystem → Software Catalog**
3. Type **local storage** in the **Filter by keyword** box to find the **Local Storage Operator** from the list of operators, and click on it.
4. Set the following options on the **Install Operator** page:
   a. Update channel as **stable**.
   b. Installation mode as **A specific namespace on the cluster**.
   c. Installed Namespace as **Operator recommended namespace openshift-local-storage**.
   d. Update approval as **Automatic**.
5. Click **Install**.

Verification

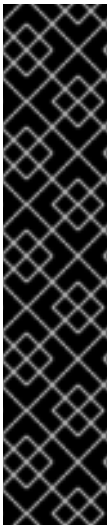
- Verify that the Local Storage Operator shows a green tick indicating successful installation.

3.2. INSTALLING RED HAT OPENSHIFT DATA FOUNDATION OPERATOR

You can install Red Hat OpenShift Data Foundation Operator using the Red Hat OpenShift Container Platform Software Catalog.

Prerequisites

- Access to an OpenShift Container Platform cluster using an account with **cluster-admin** and operator installation permissions.
- You must have at least three worker or infrastructure nodes in the Red Hat OpenShift Container Platform cluster.
- For additional resource requirements, see the Planning your deployment guide.



IMPORTANT

- When you need to override the cluster-wide default node selector for OpenShift Data Foundation, you can use the following command to specify a blank node selector for the **openshift-storage** namespace (create **openshift-storage** namespace in this case):

```
$ oc annotate namespace openshift-storage openshift.io/node-selector=
```

- Taint a node as **infra** to ensure only Red Hat OpenShift Data Foundation resources are scheduled on that node. This helps you save on subscription costs. For more information, see the *How to use dedicated worker nodes for Red Hat OpenShift Data Foundation* section in the Managing and Allocating Storage Resources guide.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Ecosystem → Software Catalog**
3. Scroll or type **OpenShift Data Foundation** into the **Filter by keyword** box to find the **OpenShift Data Foundation** Operator.
4. Click **Install**.
5. Set the following options on the **Install Operator** page:
   a. Update Channel as **stable-4.22**.
   b. Installation Mode as **A specific namespace on the cluster**
   c. Installed Namespace as **Operator recommended namespace openshift-storage**. If Namespace **openshift-storage** does not exist, it is created during the operator installation.
   d. Select Approval Strategy as **Automatic** or **Manual**.
   If you select **Automatic** updates, then the Operator Lifecycle Manager (OLM) automatically upgrades the running instance of your Operator without any intervention.

   If you select **Manual** updates, then the OLM creates an update request. As a cluster administrator, you must then manually approve that update request to update the Operator to a newer version.
   e. Ensure that the **Enable** option is selected for the **Console plugin**.
   f. Click **Install**.

Verification

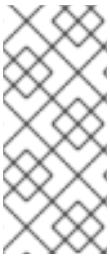
- After the operator is successfully installed, the web console automatically reloads to apply the changes. During this process, a temporary error message might appear on the page and this is expected and disappears after the refresh completes.
- In the Web Console:
  - Navigate to Installed Operators and verify that the **OpenShift Data Foundation** Operator shows a green tick indicating successful installation.
  - Navigate to **Storage** and verify if the **Data Foundation** dashboard is available.

3.3. CREATING OPENSHIFT DATA FOUNDATION CLUSTER ON VMWARE VSPHERE

Create an OpenShift Data Foundation storage cluster on VMware vSphere using the OpenShift Web Console.

VMware vSphere supports the following three types of local storage:

- Virtual machine disk (VMDK)
- Raw device mapping (RDM)
- VMDirectPath I/O



NOTE

When using thin-provisioned VMDKs with the Local Storage Operator (LSO) on VMware vSphere, all the disk space allocated to those VMDKs appears as fully consumed in the VMware vSphere interface as soon as the disks are claimed by LSO to create OSD volumes. This is expected behavior. The thin-provisioning concept at the VMware layer does not apply once OpenShift Data Foundation takes ownership of the disks.

Prerequisites

- Ensure that all the requirements in the Requirements for installing OpenShift Data Foundation using local storage devices section are met.
- You must have a minimum of three worker nodes with the same storage type and size attached to each node to use local storage devices on VMware.
- Ensure that the disk type is SSD, which is the only supported disk type.
- For VMs on VMware vSphere, ensure the **disk.EnableUUID** option is set to **TRUE**. You need to have vCenter account privileges to configure the VMs. For more information, see Required vCenter account privileges. To set the **disk.EnableUUID** option, use the **Advanced** option of the **VM Options** in the **Customize hardware** tab. For more information, see Installing on vSphere.

Procedure

1. In the OpenShift Web Console, click **Storage → Data Foundation → Storage Systems → Create StorageSystem**.

2. In the **Backing storage** page, select the **Create a new StorageClass using the local storage devices** option and click **Next**.

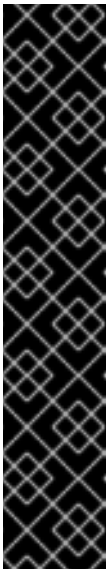
3. In the **Optional settings** page, you can configure the following optional items:

   - **Enable Network Files System**
     Automatically enable NFS for the cluster

   - **Use Ceph RBD as the default StorageClass**
     Sets Ceph RBD as the default StorageClass, eliminating the need to manually annotate a StorageClass.

   - **Set default StorageClass for virtualization**
     Marks the RBD virtualization storage class as the default for KubeVirt VM disks (PVs) after installation.

   - **Use external PostgreSQL** [Technology preview]
     Enables the use of an external PostgreSQL instance. This provides a high-availability solution for the Multicloud Object Gateway, where the PostgreSQL pod is a single point of failure.

     IMPORTANT

     OpenShift Data Foundation ships PostgreSQL images maintained by Red Hat, which are used to store metadata for the Multicloud Object Gateway. This PostgreSQL usage is at the application level.

     As a result, OpenShift Data Foundation does not perform database-level optimizations or in-depth insights.

     If you have well-maintained and optimized PostgreSQL instance, then it is recommended to use it. OpenShift Data Foundation supports external PostgreSQL instances.

     Any PostgreSQL-related issues requiring code changes or deep technical analysis may need to be addressed upstream. This could result in longer resolution times.

     a. Provide the following connection details:

        - **Username**
        - **Password**
        - **Server name** and **Port**
        - **Database name**

     b. Select **Enable TLS/SSL** checkbox to enable encryption for the Postgres server.

   - **Enable automatic backup**
     Allows automatic backup for the Multicloud Object Gateway metadata database.

     a. Select the backup frequency: Daily, Weekly, or Monthly.

     b. Specify the number of backups to retain.



NOTE

This option is disabled if the **Use external PostgreSQL** option is selected.

c. Click **Next**.

4. In the **Create local volume set** page, provide the following information:

a. Enter a name for the **LocalVolumeSet** and the **StorageClass**.
By default, the local volume set name appears for the storage class name. You can change the name.

b. Select one of the following:

- **Disks on all nodes** to use the available disks that match the selected filters on all nodes.
- **Disks on selected nodes** to use the available disks that match the selected filters only on selected nodes.



IMPORTANT

- The flexible scaling feature is enabled only when the storage cluster that you created with 3 or more nodes is spread across fewer than the minimum requirement of 3 availability zones.
  For information about flexible scaling, see knowledgebase article on *Scaling OpenShift Data Foundation cluster using YAML when flexible scaling is enabled*.
- Flexible scaling features get enabled at the time of deployment and can not be enabled or disabled later on.
- If the nodes selected do not match the OpenShift Data Foundation cluster requirement of an aggregated 30 CPUs and 72 GiB of RAM, a minimal cluster is deployed.
  For minimum starting node requirements, see the Resource requirements section in the Planning guide.

c. From the available list of **Disk Type**, select **SSD/NVMe**.

d. Expand the **Advanced** section and set the following options:

Volume Mode	Block is selected by default.
Device Type	Select one or more device types from the dropdown list.
Disk Size	Set a minimum size of 100GB for the device and maximum available size of the device that needs to be included.
Maximum Disks Limit	This indicates the maximum number of PVs that can be created on a node. If this field is left empty, then PVs are created for all the available disks on the matching nodes.

e. Click **Next**.
A pop-up to confirm the creation of LocalVolumeSet is displayed.

f. Click **Yes** to continue.

5. In the **Capacity and nodes** page, configure the following:

a. **Available raw capacity** is populated with the capacity value based on all the attached disks associated with the storage class. This takes some time to show up. The **Selected nodes** list shows the nodes based on the storage class.

b. In the **Configure performance** section, select one of the following performance profiles:

- Lean
Use this in a resource constrained environment with minimum resources that are lower than the recommended. This profile minimizes resource consumption by allocating fewer CPUs and less memory.

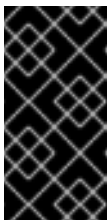
- Balanced (default)
Use this when recommended resources are available. This profile provides a balance between resource consumption and performance for diverse workloads.

- Performance
Use this in an environment with sufficient resources to get the best performance. This profile is tailored for high performance by allocating ample memory and CPUs to ensure optimal execution of demanding workloads.



NOTE

You have the option to configure the performance profile even after the deployment using the **Configure performance** option from the options menu of the **StorageSystems** tab.



IMPORTANT

Before selecting a resource profile, make sure to check the current availability of resources within the cluster. Opting for a higher resource profile in a cluster with insufficient resources might lead to installation failures.

For more information about resource requirements, see Resource requirement for performance profiles.

c. Optional: Select the **Taint nodes** checkbox to dedicate the selected nodes for OpenShift Data Foundation.

d. Click **Next**.

6. Optional: In the **Advanced settings** page, configure the following based on your requirement:

a. Select **Enable forceful deployment** to override and completely wipe any data on disks used to create this storage cluster and type **CONFIRM** in the text box.



WARNING

Any data on the disks used to create this storage cluster will be deleted. This cannot be undone. Use this option only when you are certain that the disks do not contain any data you need to preserve, such as when reusing disks from a previous OpenShift Data Foundation installation or when disks contain data from other Ceph clusters.

b. Select **Use erasure coding** to enable erasure coding instead of replication to protect the cluster data.
Erasure coding is supported for Block (RBD), File (CephFS), and RADOS Gateway (RGW) object storage.

For more information, see Erasure coding considerations.

a. Click **Next**.

1. Optional: In the **Security and network** page, configure the following based on your requirement:

b. To enable encryption, select **Enable data encryption for block and file storage**

c. Select one of the following **Encryption level**:

- **Cluster-wide encryption** to encrypt the entire cluster (block and file).
- **StorageClass encryption** to create encrypted persistent volume (block only) using encryption enabled storage class.

d. Optional: Select the **Connect to an external key management service** checkbox. This is optional for cluster-wide encryption.

i. From the **Key Management Service Provider** drop-down list, select one of the following providers and provide the necessary details:

- **Vault**

A. Select an **Authentication Method**.

- **Using Token authentication method**
  - Enter a unique **Connection Name**, host **Address** of the Vault server ('https://<hostname or ip>'), **Port** number and **Token**.
  - Expand **Advanced Settings** to enter additional settings and certificate details based on your **Vault** configuration:
    - Enter the Key Value secret path in **Backend Path** that is dedicated and unique to OpenShift Data Foundation.
  - Optional: Enter **TLS Server Name** and **Vault Enterprise Namespace**.
  - Upload the respective PEM encoded certificate file to provide the **CA Certificate**, **Client Certificate** and **Client Private Key** .

- Click **Save**.

- **Using Kubernetes authentication method**

  - Enter a unique Vault **Connection Name**, host **Address** of the Vault server ('https://<hostname or ip>'), **Port** number and **Role** name.

  - Expand **Advanced Settings** to enter additional settings and certificate details based on your **Vault** configuration:

    - Enter the Key Value secret path in **Backend Path** that is dedicated and unique to OpenShift Data Foundation.

    - Optional: Enter **TLS Server Name** and **Authentication Path** if applicable.

    - Upload the respective PEM encoded certificate file to provide the **CA Certificate**, **Client Certificate** and **Client Private Key**.

  - Click **Save**.



NOTE

In case you need to enable key rotation for Vault KMS, run the following command in the OpenShift web console after the storage cluster is created:

```
oc patch storagecluster ocs-storagecluster -n openshift-storage --type=json -p '[{"op": "add", "path":"/spec/encryption/keyRotation/enable", "value": true}]'
```

- **Thales CipherTrust Manager (using KMIP)**

  A. Enter a unique **Connection Name** for the Key Management service within the project.

  B. In the **Address** and **Port** sections, enter the IP of Thales CipherTrust Manager and the port where the KMIP interface is enabled. For example:

     - **Address**: 123.34.3.2

     - **Port**: 5696

  C. Upload the **Client Certificate**, **CA certificate**, and **Client Private Key**.

  D. If StorageClass encryption is enabled, enter the Unique Identifier to be used for encryption and decryption generated above.

  E. The **TLS Server** field is optional and used when there is no DNS entry for the KMIP endpoint. For example, **kmip_all_<port>.ciphertrustmanager.local**.

ii. Select a **Network**.

e. Click **Next**.

1. In the **Review and create** page, review the configuration details.

- To modify any configuration settings, click **Back** to go back to the previous configuration page.

2. Click **Create StorageSystem**.



NOTE

When your deployment has five or more nodes, racks, or rooms, and when there are five or more number of failure domains present in the deployment, you can configure Ceph monitor counts based on the number of racks or zones. An alert is displayed in the notification panel or Alert Center of the OpenShift Web Console to indicate the option to increase the number of Ceph monitor counts. You can use the **Configure** option in the alert to configure the Ceph monitor counts. For more information, see Resolving low Ceph monitor count alert .

Verification

- To verify the final Status of the installed storage cluster:

  a. In the OpenShift Web Console, navigate to **Storage → Data Foundation → Storage System → ocs-storagecluster**.

  b. Verify that the **Status** of **StorageCluster** is **Ready** and has a green tick mark next to it.

- To verify if flexible scaling is enabled on your storage cluster, perform the following steps (for arbiter mode, flexible scaling is disabled):

  1. In the OpenShift Web Console, navigate to **Storage → Data Foundation → Storage System**.

  2. From the Action menu, select **Edit Storage System**.

  3. In the YAML tab, search for the keys **flexibleScaling** in **spec** section and **failureDomain** in **status** section. If **flexible scaling** is true and **failureDomain** is set to host, the flexible scaling feature is enabled.

```
spec:
flexibleScaling: true
[…]
status:
failureDomain: host
```

- To verify that all components for OpenShift Data Foundation are successfully installed, see Verifying your OpenShift Data Foundation deployment .

Additional resources

- Scaling Storage guide

CHAPTER 4. VERIFYING OPENSHIFT DATA FOUNDATION DEPLOYMENT

Use this section to verify that OpenShift Data Foundation is deployed correctly.

4.1. VERIFYING THE STATE OF THE PODS

Verify that all required OpenShift Data Foundation pods are in a **Running** state after deployment.

Procedure

1. Click **Workloads → Pods** from the OpenShift Web Console.
2. Select **openshift-storage** from the **Project** drop-down list.

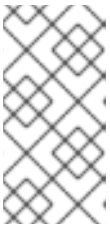


NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

For more information on the expected number of pods for each component and how it varies depending on the number of nodes, see the following table:

1. Set filter for Running and Completed pods to verify that the following pods are in **Running** and **Completed** state:



NOTE

The available pods depend on the cluster configuration. When the cluster is deployed as a standalone Multicloud Object Gateway, the **rook-ceph-operator-*** pods are not available. Similarly, when the cluster is deployed without the Multicloud Object Gateway, **noobaa-*** pods are not available.

Pods corresponding to OpenShift Data Foundation cluster

Component	Corresponding pods
OpenShift Data Foundation Operator	• **ocs-operator-*** (1 pod on any storage node) • **ocs-metrics-exporter-*** (1 pod on any storage node) • **odf-operator-controller-manager-*** (1 pod on any storage node) • **odf-console-*** (1 pod on any storage node) • **csi-addons-controller-manager-*** (1 pod on any storage node)

Rook-ceph Operator	**rook-ceph-operator-*** (1 pod on any storage node)
Multicloud Object Gateway	• **noobaa-operator-*** (1 pod on any storage node) • **noobaa-core-*** (1 pod on any storage node) • **noobaa-db-pg-cluster-1** and noobaa-db-pg-cluster-2 (2 instances of MCG DB pod on any storage node) • **noobaa-endpoint-*** (1 pod on any storage node) • **cnpg-controller-manager-*** (1 pod on any storage node)
MON	**rook-ceph-mon-*** (3 pods distributed across storage nodes)
MGR	**rook-ceph-mgr-*** (2 pods on different storage nodes, one active, one standby)
MDS	**rook-ceph-mds-ocs-storagecluster-cephfilesystem-*** (2 pods distributed across storage nodes)
RGW	**rook-ceph-rgw-ocs-storagecluster-cephobjectstore-*** (1 pod on any storage node)

CSI	• **cephfs** ◦ **openshift-storage.cephfs.csi.ceph.com-ctrlplugin-*** (2 pods distributed across storage nodes) ◦ **openshift-storage.cephfs.csi.ceph.com-nodeplugin-*** (1 pod on each storage node) • **nfs** ◦ **openshift-storage.nfs.csi.ceph.com-ctrlplugin-*** (2 pods distributed across storage nodes) ◦ **openshift-storage.nfs.csi.ceph.com-nodeplugin-*** (1 pod on each storage node) • **rbd** ◦ **openshift-storage.rbd.csi.ceph.com-ctrlplugin-*** (2 pods distributed across storage nodes) ◦ **openshift-storage.rbd.csi.ceph.com-nodeplugin-*** (1 pod on each storage node)
rook-ceph-crashcollector	**rook-ceph-crashcollector-*** (1 pod on each storage node)
OSD	• **rook-ceph-osd-*** (1 pod for each device) • **rook-ceph-osd-prepare-ocs-*** (1 pod for each device)
odf-blackbox-exporter	**odf-blackbox-exporter-*** (1 pod on any storage node) **NOTE:** This does not apply to clusters without storage, for example, Multicloud Object Gateway-only deployments or external mode deployments.

4.2. VERIFYING THE OPENSHIFT DATA FOUNDATION CLUSTER IS HEALTHY

Verify that the OpenShift Data Foundation cluster is healthy after installation.

Procedure

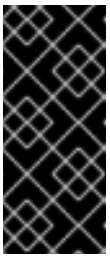
1. In the OpenShift Web Console, click **Storage → Data Foundation**.
2. In the **Status** card of the **Overview** tab, click **Storage System** and then click the storage system link from the pop up that appears.
3. In the **Status** card of the **Block and File** tab, verify that the *Storage Cluster* has a green tick.
4. In the **Details** card, verify that the cluster information is displayed.
   For more information on the health of the OpenShift Data Foundation cluster using the **Block and File** dashboard, see Monitoring OpenShift Data Foundation.

4.3. VERIFYING THE MULTICLOUD OBJECT GATEWAY IS HEALTHY

Verify that the Multicloud Object Gateway is healthy after deployment to confirm that the object service and data resiliency are operational and that the NooBaa database has a recent backup.

Procedure

1. In the OpenShift Web Console, click **Storage → Data Foundation**.
2. In the **Status** card of the **Overview** tab, click **Storage System** and then click the storage system link from the pop up that appears.
   a. In the **Status card** of the **Object** tab, verify that both *Object Service* and *Data Resiliency* have a green tick.
   b. In the **Details** card, verify that the MCG information is displayed.
   For more information on the health of the OpenShift Data Foundation cluster using the object service dashboard, see Monitoring OpenShift Data Foundation.



IMPORTANT

To avoid data loss, it is recommended to take a backup of NooBaa DB PVC regularly. If NooBaa DB fails and cannot be recovered, then you can revert to the latest backed-up version. For instructions on backing up your NooBaa DB, follow the steps in the knowledgebase article, Perform a One-Time Backup of the Database for the Multicloud Object Gateway.

4.4. VERIFYING THAT THE SPECIFIC STORAGE CLASSES EXIST

Verify that the expected OpenShift Data Foundation storage classes are present after deployment.

Procedure

1. Click **Storage → Storage Classes**from the left pane of the OpenShift Web Console.
2. Verify that the following storage classes are created with the OpenShift Data Foundation cluster creation:
   - **ocs-storagecluster-ceph-rbd**

- **ocs-storagecluster-cephfs**
- **openshift-storage.noobaa.io**
- **ocs-storagecluster-ceph-rgw**

CHAPTER 5. DEPLOY STANDALONE MULTICLOUD OBJECT GATEWAY

Deploying only the Multicloud Object Gateway component with the OpenShift Data Foundation provides the flexibility in deployment and helps to reduce the resource consumption. You can deploy the Multicloud Object Gateway component either using dynamic storage devices or using the local storage devices.

CHAPTER 6. DEPLOY STANDALONE MULTICLOUD OBJECT GATEWAY USING DYNAMIC STORAGE DEVICES

Use this section to deploy only the standalone Multicloud Object Gateway component, which involves the following steps:

- Installing Red Hat OpenShift Data Foundation Operator
- Creating standalone Multicloud Object Gateway

After deploying the component, you can create and manage buckets using MCG object browser. For more information, see Creating and managing buckets using MCG object browser .

6.1. INSTALLING RED HAT OPENSHIFT DATA FOUNDATION OPERATOR

You can install Red Hat OpenShift Data Foundation Operator using the Red Hat OpenShift Container Platform Software Catalog.

Prerequisites

- Access to an OpenShift Container Platform cluster using an account with **cluster-admin** and operator installation permissions.
- You must have at least three worker or infrastructure nodes in the Red Hat OpenShift Container Platform cluster.
- For additional resource requirements, see the Planning your deployment guide.



IMPORTANT

- When you need to override the cluster-wide default node selector for OpenShift Data Foundation, you can use the following command to specify a blank node selector for the **openshift-storage** namespace (create **openshift-storage** namespace in this case):

```
$ oc annotate namespace openshift-storage openshift.io/node-selector=
```

- Taint a node as **infra** to ensure only Red Hat OpenShift Data Foundation resources are scheduled on that node. This helps you save on subscription costs. For more information, see the *How to use dedicated worker nodes for Red Hat OpenShift Data Foundation* section in the Managing and Allocating Storage Resources guide.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Ecosystem → Software Catalog**
3. Scroll or type **OpenShift Data Foundation** into the **Filter by keyword** box to find the **OpenShift Data Foundation** Operator.

4. Click **Install**.
5. Set the following options on the **Install Operator** page:
    a. Update Channel as **stable-4.22**.
    b. Installation Mode as **A specific namespace on the cluster**.
    c. Installed Namespace as **Operator recommended namespace openshift-storage**. If Namespace **openshift-storage** does not exist, it is created during the operator installation.
    d. Select Approval Strategy as **Automatic** or **Manual**.
    If you select **Automatic** updates, then the Operator Lifecycle Manager (OLM) automatically upgrades the running instance of your Operator without any intervention.

    If you select **Manual** updates, then the OLM creates an update request. As a cluster administrator, you must then manually approve that update request to update the Operator to a newer version.
    e. Ensure that the **Enable** option is selected for the **Console plugin**.
    f. Click **Install**.

Verification

- After the operator is successfully installed, the web console automatically reloads to apply the changes. During this process, a temporary error message might appear on the page and this is expected and disappears after the refresh completes.
- In the Web Console:
    - Navigate to Installed Operators and verify that the **OpenShift Data Foundation** Operator shows a green tick indicating successful installation.
    - Navigate to **Storage** and verify if the **Data Foundation** dashboard is available.

6.2. CREATING A STANDALONE MULTICLOUD OBJECT GATEWAY

You can create only the standalone Multicloud Object Gateway (MCG) component while deploying OpenShift Data Foundation. After you create the MCG component, you can create and manage buckets using the MCG object browser. For more information, see Creating and managing buckets using MCG object browser.

Prerequisites

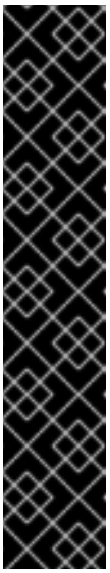
- Ensure that the OpenShift Data Foundation Operator is installed.
- The Multicloud Object Gateway database uses ReadWriteOncePod (RWOP) access mode to safeguard against node failures. If the underlying CSI driver does not support RWOP, the access mode must be manually changed to ReadWriteOnce (RWO) before deployment. For more information, see the knowledgebase article How to change NooBaa DB access mode from RWOP to RWO in standalone MCG deployments?.

Procedure

1. In the OpenShift Web Console, navigate to **Storage → StorageCluster**.

2. On the Welcome page, click Configure Data Foundation.
3. Select Setup Multicloud Object Gateway.
4. In the **Backing storage** page, select the following:
    a. Select the **Use an existing StorageClass** option.
    b. Click **Next**.
5. In the **Advanced Settings** page, you can configure the following optional items:

- **Use external PostgreSQL** [Technology preview]
  Enables the use of an external PostgreSQL instance. This provides a high-availability solution for the Multicloud Object Gateway, where the PostgreSQL pod is a single point of failure.



IMPORTANT

OpenShift Data Foundation ships PostgreSQL images maintained by Red Hat, which are used to store metadata for the Multicloud Object Gateway. This PostgreSQL usage is at the application level.

As a result, OpenShift Data Foundation does not perform database-level optimizations or in-depth insights.

If you have well-maintained and optimized PostgreSQL instance, then it is recommended to use it. OpenShift Data Foundation supports external PostgreSQL instances.

Any PostgreSQL-related issues requiring code changes or deep technical analysis may need to be addressed upstream. This could result in longer resolution times.

    a. Provide the following connection details:
        - **Username**
        - **Password**
        - **Server name** and **Port**
        - **Database name**
    b. Select **Enable TLS/SSL** checkbox to enable encryption for the Postgres server.

- **Enable automatic backup**
  Allows automatic backup for the NooBaa database.
    a. Select the backup frequency: Daily, Weekly, or Monthly.
    b. Specify the number of backups to retain.
    c. Select a VolumeSnapshotClass to add a volume snapshot.



NOTE

This option is disabled if the **Use external PostgreSQL** option is selected.

d. Click **Next**.

6. In the **Security** page, select any of the following optional items:

   a. Select the **Connect to an external key management service** checkbox. This is optional for cluster-wide encryption.

   b. From the **Key Management Service Provider** drop-down list, either select **Vault** or **Thales CipherTrust Manager (using KMIP)**. If you selected **Vault**, go to the next step. If you selected **Thales CipherTrust Manager (using KMIP)**, go to step iii.

   c. Select an **Authentication Method**.

      Using Token authentication method

      - Enter a unique **Connection Name**, host **Address** of the Vault server ('https://<hostname or ip>'), **Port** number and **Token**.
      - Expand **Advanced Settings** to enter additional settings and certificate details based on your **Vault** configuration:
        - Enter the Key Value secret path in **Backend Path** that is dedicated and unique to OpenShift Data Foundation.
        - Optional: Enter **TLS Server Name**, **Authentication Path**, and **Vault Enterprise Namespace**.
        - Upload the respective PEM encoded certificate file to provide the **CA Certificate**, **Client Certificate** and **Client Private Key** .
        - Click **Save** and skip to step iv.

      Using Kubernetes authentication method

      - Enter a unique Vault **Connection Name**, host **Address** of the Vault server ('https://<hostname or ip>'), **Port** number and **Role** name.
      - Expand **Advanced Settings** to enter additional settings and certificate details based on your **Vault** configuration:
        - Enter the Key Value secret path in **Backend Path** that is dedicated and unique to OpenShift Data Foundation.
        - Optional: Enter **TLS Server Name**, **Authentication Path**, and **Vault Enterprise Namespace** if applicable.
        - Upload the respective PEM encoded certificate file to provide the **CA Certificate**, **Client Certificate** and **Client Private Key** .
        - Click **Save** and skip to step iv.

d. To use **Thales CipherTrust Manager (using KMIP)** as the KMS provider, follow the steps below:

   i. Enter a unique **Connection Name** for the Key Management service within the project.

   ii. In the **Address** and **Port** sections, enter the IP of Thales CipherTrust Manager and the port where the KMIP interface is enabled. For example:

   - **Address**: 123.34.3.2
   - **Port**: 5696

   iii. Upload the **Client Certificate**, **CA certificate**, and **Client Private Key**.

   iv. If StorageClass encryption is enabled, enter the Unique Identifier to be used for encryption and decryption generated above.

   v. The **TLS Server** field is optional and used when there is no DNS entry for the KMIP endpoint. For example, **kmip_all_<port>.ciphertrustmanager.local**.

e. Select a **Network**.

f. Click **Next**.

7. In the **Review and create** page, review the configuration details:
   To modify any configuration settings, click **Back**.

8. Click **Create StorageSystem**.

Verification

Verifying that the OpenShift Data Foundation cluster is healthy

1. In the OpenShift Web Console, click **Storage → Data Foundation**.

2. In the **Status** card of the **Overview** tab, click **Storage System** and then click the storage system link from the pop up that appears.

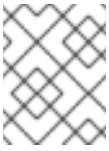
   a. In the **Status card** of the **Object** tab, verify that both *Object Service* and *Data Resiliency* have a green tick.

   b. In the **Details** card, verify that the MCG information is displayed.

Verifying the state of the pods

1. Click **Workloads → Pods** from the OpenShift Web Console.

2. Select **openshift-storage** from the **Project** drop-down list and verify that the following pods are in **Running** state.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

Component	Corresponding pods
OpenShift Data Foundation Operator	• **ocs-operator-*** (1 pod on any storage node) • **ocs-metrics-exporter-*** (1 pod on any storage node) • **odf-operator-controller-manager-*** (1 pod on any storage node) • **odf-console-*** (1 pod on any storage node) • **csi-addons-controller-manager-*** (1 pod on any storage node)
Rook-ceph Operator	**rook-ceph-operator-*** (1 pod on any storage node)
Multicloud Object Gateway	• **noobaa-operator-*** (1 pod on any storage node) • **noobaa-core-*** (1 pod on any storage node) • **noobaa-db-pg-cluster-1** and noobaa-db-pg-cluster-2 (2 instances of MCG DB pod on any storage node) • **noobaa-endpoint-*** (1 pod on any storage node) • **cnpg-controller-manager-*** (1 pod on any storage node)

CHAPTER 7. DEPLOY STANDALONE MULTICLOUD OBJECT GATEWAY USING LOCAL STORAGE DEVICES

Use this section to deploy only the standalone Multicloud Object Gateway component, which involves the following steps:

- Installing the Local Storage Operator
- Installing Red Hat OpenShift Data Foundation Operator
- Creating standalone Multicloud Object Gateway

After deploying the MCG component, you can create and manage buckets using MCG object browser. For more information, see Creating and managing buckets using MCG object browser .

7.1. INSTALLING LOCAL STORAGE OPERATOR

Install the Local Storage Operator from the Software Catalog before creating Red Hat OpenShift Data Foundation clusters on local storage devices.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Ecosystem → Software Catalog**
3. Type **local storage** in the **Filter by keyword** box to find the **Local Storage Operator** from the list of operators, and click on it.
4. Set the following options on the **Install Operator** page:
    a. Update channel as **stable**.
    b. Installation mode as **A specific namespace on the cluster**
    c. Installed Namespace as **Operator recommended namespace openshift-local-storage**.
    d. Update approval as **Automatic**.
5. Click **Install**.

Verification

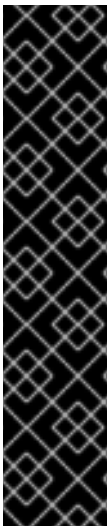
- Verify that the Local Storage Operator shows a green tick indicating successful installation.

7.2. INSTALLING RED HAT OPENSHIFT DATA FOUNDATION OPERATOR

You can install Red Hat OpenShift Data Foundation Operator using the Red Hat OpenShift Container Platform Software Catalog.

Prerequisites

- Access to an OpenShift Container Platform cluster using an account with **cluster-admin** and operator installation permissions.
- You must have at least three worker or infrastructure nodes in the Red Hat OpenShift Container Platform cluster.
- For additional resource requirements, see the Planning your deployment guide.



IMPORTANT

- When you need to override the cluster-wide default node selector for OpenShift Data Foundation, you can use the following command to specify a blank node selector for the **openshift-storage** namespace (create **openshift-storage** namespace in this case):

```
$ oc annotate namespace openshift-storage openshift.io/node-selector=
```

- Taint a node as **infra** to ensure only Red Hat OpenShift Data Foundation resources are scheduled on that node. This helps you save on subscription costs. For more information, see the *How to use dedicated worker nodes for Red Hat OpenShift Data Foundation* section in the Managing and Allocating Storage Resources guide.

Procedure

1. Log in to the OpenShift Web Console.
2. Click **Ecosystem → Software Catalog**
3. Scroll or type **OpenShift Data Foundation** into the **Filter by keyword** box to find the **OpenShift Data Foundation** Operator.
4. Click **Install**.
5. Set the following options on the **Install Operator** page:
   a. Update Channel as **stable-4.22**.
   b. Installation Mode as **A specific namespace on the cluster**
   c. Installed Namespace as **Operator recommended namespace openshift-storage**. If Namespace **openshift-storage** does not exist, it is created during the operator installation.
   d. Select Approval Strategy as **Automatic** or **Manual**.
   If you select **Automatic** updates, then the Operator Lifecycle Manager (OLM) automatically upgrades the running instance of your Operator without any intervention.

   If you select **Manual** updates, then the OLM creates an update request. As a cluster administrator, you must then manually approve that update request to update the Operator to a newer version.
   e. Ensure that the **Enable** option is selected for the **Console plugin**.
   f. Click **Install**.

Verification

- After the operator is successfully installed, the web console automatically reloads to apply the changes. During this process, a temporary error message might appear on the page and this is expected and disappears after the refresh completes.
- In the Web Console:
  - Navigate to Installed Operators and verify that the **OpenShift Data Foundation** Operator shows a green tick indicating successful installation.
  - Navigate to **Storage** and verify if the **Data Foundation** dashboard is available.

7.3. CREATING A STANDALONE MULTICLOUD OBJECT GATEWAY

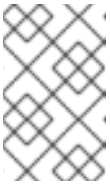
You can create only the standalone Multicloud Object Gateway (MCG) component while deploying OpenShift Data Foundation. After you create the MCG component, you can create and manage buckets using the MCG object browser. For more information, see Creating and managing buckets using MCG object browser.

Prerequisites

- Ensure that the OpenShift Data Foundation Operator is installed.
- The Multicloud Object Gateway database uses ReadWriteOncePod (RWOP) access mode to safeguard against node failures. If the underlying CSI driver does not support RWOP, the access mode must be manually changed to ReadWriteOnce (RWO) before deployment. For more information, see the knowledgebase article How to change NooBaa DB access mode from RWOP to RWO in standalone MCG deployments?.

Procedure

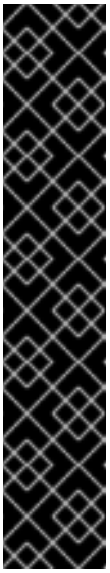
1. In the OpenShift Web Console, navigate to **Storage → StorageCluster**.
2. On the Welcome page, click Configure Data Foundation.
3. Select Setup Multicloud Object Gateway.
4. In the **Backing storage** page, select the following:
   a. Select the **Create a new StorageClass using the local storage devices**option.
   b. Click **Next**.



NOTE

You are prompted to install the Local Storage Operator if it is not already installed. Click **Install**, and follow the procedure as described in Installing Local Storage Operator.

5. In the **Advanced Settings** page, you can configure the following optional items:
   - **Use external PostgreSQL** [Technology preview]
     Enables the use of an external PostgreSQL instance. This provides a high-availability solution for the Multicloud Object Gateway, where the PostgreSQL pod is a single point of failure.



IMPORTANT

OpenShift Data Foundation ships PostgreSQL images maintained by Red Hat, which are used to store metadata for the Multicloud Object Gateway. This PostgreSQL usage is at the application level.

As a result, OpenShift Data Foundation does not perform database-level optimizations or in-depth insights.

If you have well-maintained and optimized PostgreSQL instance, then it is recommended to use it. OpenShift Data Foundation supports external PostgreSQL instances.

Any PostgreSQL-related issues requiring code changes or deep technical analysis may need to be addressed upstream. This could result in longer resolution times.

a. Provide the following connection details:
   - **Username**
   - **Password**
   - **Server name** and **Port**
   - **Database name**

b. Select **Enable TLS/SSL** checkbox to enable encryption for the Postgres server.

- **Enable automatic backup**
  Allows automatic backup for the NooBaa database.

  a. Select the backup frequency: Daily, Weekly, or Monthly.

  b. Specify the number of backups to retain.

  c. Select a VolumeSnapshotClass to add a volume snapshot.



NOTE

This option is disabled if the **Use external PostgreSQL** option is selected.

  d. Click **Next**.

6. In the **Create local volume set** page, provide the following information:

   a. Enter a name for the **LocalVolumeSet** and the **StorageClass**.
   By default, the local volume set name appears for the storage class name. You can change the name.

   b. Choose one of the following:

   - **Disks on all nodes**
     Uses the available disks that match the selected filters on all the nodes.

- **Disks on selected nodes**
  Uses the available disks that match the selected filters only on the selected nodes.

c. From the available list of **Disk Type**, select **SSD/NVMe**.

d. Expand the **Advanced** section and set the following options:

Volume Mode	Filesystem is selected by default. Always ensure that the Filesystem is selected for **Volume Mode**.
Device Type	Select one or more device types from the dropdown list.
Disk Size	Set a minimum size of 100GB for the device and maximum available size of the device that needs to be included.
Maximum Disks Limit	This indicates the maximum number of PVs that can be created on a node. If this field is left empty, then PVs are created for all the available disks on the matching nodes.

e. Click **Next**.
A pop-up to confirm the creation of LocalVolumeSet is displayed.

f. Click **Yes** to continue.

7. In the **Capacity and nodes** page, configure the following:

   a. **Available raw capacity** is populated with the capacity value based on all the attached disks associated with the storage class. This takes some time to show up. The **Selected nodes** list shows the nodes based on the storage class.

   b. Click **Next**.

8. In the **Security** page, select any of the following optional items:

   a. Select the **Connect to an external key management service** checkbox. This is optional for cluster-wide encryption.

   b. From the **Key Management Service Provider** drop-down list, either select **Vault** or **Thales CipherTrust Manager (using KMIP)**. If you selected **Vault**, go to the next step. If you selected **Thales CipherTrust Manager (using KMIP)**, go to step iii.

   c. Select an **Authentication Method**.

   Using Token authentication method

   - Enter a unique **Connection Name**, host **Address** of the Vault server ('https://<hostname or ip>'), **Port** number and **Token**.
   - Expand **Advanced Settings** to enter additional settings and certificate details based on your **Vault** configuration:
     - Enter the Key Value secret path in **Backend Path** that is dedicated and unique to OpenShift Data Foundation.

- Optional: Enter **TLS Server Name**, **Authentication Path**, and **Vault Enterprise Namespace**.
- Upload the respective PEM encoded certificate file to provide the **CA Certificate**, **Client Certificate** and **Client Private Key** .
- Click **Save** and skip to step iv.

Using Kubernetes authentication method

- Enter a unique Vault **Connection Name**, host **Address** of the Vault server ('https://<hostname or ip>'), **Port** number and **Role** name.
- Expand **Advanced Settings** to enter additional settings and certificate details based on your **Vault** configuration:
  - Enter the Key Value secret path in **Backend Path** that is dedicated and unique to OpenShift Data Foundation.
  - Optional: Enter **TLS Server Name**, **Authentication Path**, and **Vault Enterprise Namespace** if applicable.
  - Upload the respective PEM encoded certificate file to provide the **CA Certificate**, **Client Certificate** and **Client Private Key** .
  - Click **Save** and skip to step iv.

d. To use **Thales CipherTrust Manager (using KMIP)** as the KMS provider, follow the steps below:

   i. Enter a unique **Connection Name** for the Key Management service within the project.

   ii. In the **Address** and **Port** sections, enter the IP of Thales CipherTrust Manager and the port where the KMIP interface is enabled. For example:

   - **Address**: 123.34.3.2
   - **Port**: 5696

   iii. Upload the **Client Certificate**, **CA certificate**, and **Client Private Key**.

   iv. If StorageClass encryption is enabled, enter the Unique Identifier to be used for encryption and decryption generated above.

   v. The **TLS Server** field is optional and used when there is no DNS entry for the KMIP endpoint. For example, **kmip_all_<port>.ciphertrustmanager.local**.

e. Select a **Network**.

f. Click **Next**.

9. In the **Review and create** page, review the configuration details:
   To modify any configuration settings, click **Back**.
10. Click **Create StorageSystem**.

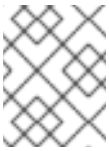
Verification

Verifying that the OpenShift Data Foundation cluster is healthy

1. In the OpenShift Web Console, click **Storage → Data Foundation**.
2. In the **Status** card of the **Overview** tab, click **Storage System** and then click the storage system link from the pop up that appears.
   a. In the **Status card** of the **Object** tab, verify that both *Object Service* and *Data Resiliency* have a green tick.
   b. In the **Details** card, verify that the MCG information is displayed.

Verifying the state of the pods

1. Click **Workloads → Pods** from the OpenShift Web Console.
2. Select **openshift-storage** from the **Project** drop-down list and verify that the following pods are in **Running** state.



NOTE

If the **Show default projects** option is disabled, use the toggle button to list all the default projects.

Component	Corresponding pods
OpenShift Data Foundation Operator	• **ocs-operator-*** (1 pod on any storage node) • **ocs-metrics-exporter-*** (1 pod on any storage node) • **odf-operator-controller-manager-*** (1 pod on any storage node) • **odf-console-*** (1 pod on any storage node) • **csi-addons-controller-manager-*** (1 pod on any storage node)
Rook-ceph Operator	**rook-ceph-operator-*** (1 pod on any storage node)

Component	Corresponding pods
Multicloud Object Gateway	• **noobaa-operator-*** (1 pod on any storage node) • **noobaa-core-*** (1 pod on any storage node) • **noobaa-db-pg-cluster-1** and noobaa-db-pg-cluster-2 (2 instances of MCG DB pod on any storage node) • **noobaa-endpoint-*** (1 pod on any storage node) • **cnpg-controller-manager-*** (1 pod on any storage node)

CHAPTER 8. VIEW OPENSHIFT DATA FOUNDATION TOPOLOGY

The topology provides a visual representation of the OpenShift Data Foundation storage cluster at multiple levels of abstraction and enables interaction with each layer. It illustrates how the cluster components are organized and related, while offering visibility into Object Storage Devices (OSDs), including their count, type, and status.

Procedure

1. On the OpenShift Web Console, navigate to **Storage → Data Foundation → Topology**.
   The view shows the storage cluster and the topology groupings inside it. You can see the nodes depicted by circular entities within the topology groupings, which are indicated by dotted lines. The topology groupings are based on the cluster's topology configuration (for example, racks in bare metal or vSphere clusters, or zones in cloud-based clusters). The label of each item or resource contains basic information such as status and health or indication for alerts.

2. Choose a node to view node details on the right-hand panel. You can also access resources or deployments within a node by clicking on the search/preview decorator icon.
   The node details panel displays information about the OSDs running on that node, including:

   - Number of OSDs on the node
   - OSD disk type (for example, SSD, HDD)
   - OSD status (for example, Running, Error, CrashLoopBackOff)

3. To view deployment details

   a. Click the preview decorator on a node. A modal window appears above the node that displays all of the deployments associated with that node along with their statuses.

   b. Click the **Back to main view** button in the model's upper left corner to close and return to the previous view.

   c. Select a specific deployment to see more information about it. All relevant data is shown in the side panel.

4. Click the **Resources** tab to view the pods information. This tab provides a deeper understanding of the problems and offers granularity that aids in better troubleshooting.

5. Click the **OSD information** tab to view detailed information about the OSDs. This tab provides more comprehensive OSD-specific details, including:

   - OSD pod name and status
   - Disk device associated with each OSD
   - Device class (for example, SSD, HDD)
   - Resource utilization metrics

6. Click the pod links to view the pod information page on OpenShift Container Platform. The link opens in a new window.

CHAPTER 9. UNINSTALLING OPENSHIFT DATA FOUNDATION

9.1. UNINSTALLING OPENSHIFT DATA FOUNDATION IN INTERNAL MODE

Uninstalling OpenShift Data Foundation in internal mode permanently removes the storage cluster, all associated PVCs, and the operator.

Procedure

1. To uninstall OpenShift Data Foundation in internal mode, refer to the knowledgebase article on Uninstalling OpenShift Data Foundation.